

1 版本信息【Version Information】

版本【Version】	时间【Date】	说明【Description】
V1.0	2019-12-12	初始版本
V1.1	2020-02-11	调整黑体设定温度、增加人体测温标定补充
V1.2	2020-02-25	新增二次标定单点标定功能
V2.0	2020-08-22	修改二次标定功能相关的命令接口为新的SDK接口
V3.0	2021-05-17	适配ASIC相关指令

2 概述【Overview】

Tiny红外模组集成到整机中后，需要测温二次标定才可获得较准确的测温效果。本文系统的介绍了二次标定实现流程。测温二次标定计算在模组固件中已经实现，上位机只需调用libiruvc库中的相关命令接口按照标定流程操作即可完成标定过程。相关的命令用法参考SDK中libirsdk.chm接口说明文档。

【When the user integrates the module into the whole machine, it will inevitably cause a certain temperature measurement deviation due to the changes of thermal distribution and optical structure (add window, etc.). Therefore, Secondary calibration& Lid pattern noise correction is required in the whole machine. The secondary calibration calculation has been realized in module firmware, and the master terminal only needs to call the relevant command interface in libiruvc library to complete the calibration process. Refer to the libirsdk. chm interface description in the SDK. 】

3 两点标定【Two-point secondary calibration】

备注：在所有的标定之前(包含盲元标定，测温二次标定，锅盖标定)请先调用 IRCMD 类中 `restoreDefaultConfig` 接口的 `DEF_CFG_TPD` 方法，以重置标定数据，提高标定的准确性。

【Remarks: Before all calibrations (including blind element calibration, temperature measurement secondary calibration, pot lid calibration), please call the `DEF_CFG_TPD` method of the `restoreDefaultConfig` interface in the `IRCMD` class to reset the calibration data and improve the Calibration accuracy. 】

请注意：调用该方法之后，会重置所有的标定数据。【Note: After calling this method, all calibration data will be reset.】

即如果盲元标定结束之后，在锅盖标定的时候，调用了该方法，则盲元标定的数据会丢失。

【That is, if this method is called when the lid is calibrated after the blind element calibration is completed, the data of the blind element calibration will be lost.】

3.1 准备工作【Preparation】

3.1.1 Step1

准备两台黑体A和B，温度设置为TA和TB；

【Prepare two black bodies A and B, and set the temperature to TA and TB.】

3.1.2 Step2

黑体分别可以放置于模组视场中心，黑体面距离模组镜头约0.25m。

【The black body can be placed in the center of the module field of view, and the black face is about 0.25m away from the module lens.】

建议高增益模式下，TA和TB分别设置为293K和393K，低增益模式下，TA和TB分别设置为373K和673K

【You are advised to set TA and TB to 293K and 393K respectively in high gain mode and 373K and 673K respectively in low gain mode】

人体测温只需要在高增益模式下标定，TA=303K，TB=313K

【Creature Radiometry only needs to be calibrated in high-gain mode, TA=303K, TB=313K】

使工业高增益模式的用户，如果对5°C以下的低温目标精度比较关心，可将第一个温度点温度降低到283K。

【Industrial high-gain mode: If you are concerned about the temperature measurement accuracy of the low-temperature target below 5°C, you can lower the temperature of the first temperature point to 10°C.】

3.1.3 Step3

等待黑体达到设定温度

【Wait for the blackbody to reach the set temperature】

实际黑体发射率不可能为1，需要修正。黑体面温度均匀稳定的区域至少需要占据画面中心9*9像素以上。

【The actual blackbody emissivity cannot be 100% and needs to be corrected. The area with uniform and stable temperature should occupy at least 9 x 9 pixels in the center of the image.】

3.2 模组温度稳定后，后进入二次标定流程【After the module temperature is stabilized, the process is entered】

3.2.1 Step1

将低温黑体A置于画面中心处

【Place the low-temperature black body A in the center of the picture】

3.2.2 Step2

调用 `setPropAutoShutterParameter` 函数的 `SHUTTER_PROP_SWITCH` 为0，关闭自动快门。

【Call `setPropAutoShutterParameter` function with `SHUTTER_PROP_SWITCH` 0 to close the auto FFC.】

```
1  /**
2   * 设置自动快门参数【Set property page's auto shutter parameters】
3   * 需要 {@link #saveSpiConfig(int)} 的 {@link
CommonParams#SPI_MOD_CFG_PROPERTY_PAGE}才保存, 否则不保存
4   *
5   * @param shutter_param param during {@link CommonParams#SHUTTER_PROP_SWITCH
自动快门开关【Automatic shutter switch】},
6   *                                {@link CommonParams#SHUTTER_PROP_MIN_INTERVAL 最小快
门间隔【Minimum shutter interval】},
7   *                                {@link CommonParams#SHUTTER_PROP_MAX_INTERVAL 最大快
门间隔【Maximum shutter interval】}
8   *                                {@link
CommonParams#SHUTTER_PROP_TEMP_THRESHOLD_OOC},
9   *                                {@link CommonParams#SHUTTER_PROP_TEMP_THRESHOLD_B 自
动快门校正触发阈值【Automatic shutter correction trigger threshold】} ,
10  * @param value
11  * @return 0 is success
12  * @see #getPropAutoShutterParameter(int)
13  * @see #saveSpiConfig(int)
14  */
15 public int setPropAutoShutterParameter(int shutter_param, int value)
```

3.2.3 Step3

调用 `shutterUpdate` 函数，进行快门校正

【Call `shutterUpdate` function for FFC】

```
1  /**
2   * 打快门【manual shutter】
3   * The firmware from 3.6 allow to shutter manually every 5 s
4   *
5   * @return 0 is success
6   */
7  public int shutterUpdate()
```

3.2.4 Step4

采集低温黑体数据，调用 `setTPDKtBtRecal2Point` 函数接口，设置第一个参数为 `TPD_KTBT_RECAL_P1`，第二个参数为黑体的实际开尔文温度 `K(integer)`。

【Collect low-temperature blackbody data, call `setTPDKtBtRecal2Point` function interface, set the first parameter as `TPD_KTBT_RECAL_P1`, and set the second parameter as the actual blackbody temperature Kelvin `K(integer)`.】

```

1  /**
2      * 两点二次标定【Recalculate the kt_bt by 2 points' temperature】
3      * First set point1(low temp),then set point2(high temp).
4      * 标定时间大约需要十几秒，可以保存【The calibration time takes about ten
seconds and can be saved】
5      *
6      * @param pointType  {@link CommonParams#TPD_KTBT_RECAL_P1} is
point1(low temp) and
7      *                    {@link CommonParams#TPD_KTBT_RECAL_P2} is
point2(high temp)
8      * @param temperature the target black body's temperature.
9      *                    units:K(integer)
10     * @return
11     */
12     public int setTPDKtBtRecal2Point(int pointType, int temperature)

```

3.2.5 Step5

将高温黑体B置于画面中心处

【Place the high-temperature black body B in the center of the picture】

3.2.6 Step6

调用 `shutterUpdate` 函数，进行快门校正

【Call `shutterUpdate` function for FFC】

```

1  /**
2      * 打快门【manual shutter】
3      * The firmware from 3.6 allow to shutter manually every 5 s
4      *
5      * @return 0 is success
6      */
7     public int shutterUpdate()

```

3.2.7 Step7

采集高温黑体数据，调用 `setTPDKtBtRecal2Point` 函数接口，设置第一个参数为 `TPD_KTBT_RECAL_P2`，第二个参数为黑体的实际开尔文温度 `K(integer)`。

USB命令接口需要等待11秒左右，I2C命令接口需要等待20s左右。

【Collect high-temperature blackbody data, call `setTPDKtBtRecal2Point` function interface, set the 2nd parameter as `TPD_KTBT_RECAL_P2`, and set the second parameter as the actual blackbody temperature Kelvin `K(integer)`.

The wait time for the USB command interface is about 11 seconds, and the wait time for the I2C command interface is about 20 seconds. 】

```

1  /**
2   * 两点二次标定【Recalculate the kt_bt by 2 points' temperature】
3   * First set point1(low temp),then set point2(high temp).
4   * 标定时间大约需要十几秒，可以保存【The calibration time takes about ten
seconds and can be saved】
5   *
6   * @param pointType  {@link CommonParams#TPD_KTBT_RECAL_P1} is
point1(low temp) and
7   *                    {@link CommonParams#TPD_KTBT_RECAL_P2} is
point2(high temp)
8   * @param temperature the target black body's temperature.
9   *                    units:K(integer)
10  * @return
11  */
12  public int setTPDKtBtRecal2Point(int pointType, int temperature)

```

3.2.8 Step8

调用 `setPropAutoShutterParameter` 函数的 `SHUTTER_PROP_SWITCH` 为1，打开自动快门。

【Call `setPropAutoShutterParameter` function with `SHUTTER_PROP_SWITCH` as 1 to open the auto FFC. 】

```

1  /**
2   * 设置自动快门参数【Set property page's auto shutter parameters】
3   * 需要 {@link #saveSpiConfig(int)} 的 {@link
CommonParams#SPI_MOD_CFG_PROPERTY_PAGE}才保存,否则不保存
4   *
5   * @param shutter_param param during {@link CommonParams#SHUTTER_PROP_SWITCH
自动快门开关【Automatic shutter switch】},
6   *                    {@link CommonParams#SHUTTER_PROP_MIN_INTERVAL 最小快
门间隔【Minimum shutter interval】},
7   *                    {@link CommonParams#SHUTTER_PROP_MAX_INTERVAL 最大快
门间隔【Maximum shutter interval】}
8   *                    {@link
CommonParams#SHUTTER_PROP_TEMP_THRESHOLD_OOC},
9   *                    {@link CommonParams#SHUTTER_PROP_TEMP_THRESHOLD_B 自
动快门校正触发阈值【Automatic shutter correction trigger threshold】} ,
10  * @param value
11  * @return 0 is success
12  * @see #getPropAutoShutterParameter(int)
13  * @see #saveSpiConfig(int)
14  */
15  public int setPropAutoShutterParameter(int shutter_param, int value)

```

3.3 注意事项【Notes】

- 标定失败后，需要调用 `restoreDefaultConfig` 接口的 `DEF_CFG_TPD` 重置温度计算参数，重新进行二次标定

【If the calibration fails, call `DEF_CFG_TPD` of `restoreDefaultConfig` to restore the calibration data and perform secondary calibration again. 】

```

1  /**
2   * 恢复出厂设置,重置CFG参数【Restore factory settings, reset CFG
parameters】
3   * 立即生效,可以保存【Effective immediately and can be saved】
4   *
5   * @param defaultCfgType {@link CommonParams#DEF_CFG_ALL 重置所有参数
【Reset all parameters】},
6   *                       {@link CommonParams#DEF_CFG_TECLESS_K},
7   *                       {@link CommonParams#DEF_CFG_TPD 重置测温参数
【Reset temperature measurement parameters】},
8   *                       {@link CommonParams#DEF_CFG_PROP_PAGE},
9   *                       {@link CommonParams#DEF_CFG_USER_CFG} ,
10  *                       {@link CommonParams#DEF_CFG_DEAD_PIXEL 重置盲元
表【Reset the blind meta table】}
11  * @return 0 is success
12  */
13  public int restoreDefaultConfig(int defaultCfgType)

```

- 若模组为可切换高低增益版本, 需要分别进行二次标定。
【 If the module is a switchable high-low gain version, perform secondary calibration in two cases. 】
- 必须在开启测温功能的情况下采集黑体数据, 这样才可以评估二次标定的效果。
【Data must be collected with temperature measurement enabled so that the effect can be evaluated. 】

4 单点标定功能【One-point secondary calibration】

备注: 在所有的标定之前(包含盲元标定, 测温二次标定, 锅盖标定)请先调用 `IRCMD` 类中 `restoreDefaultConfig` 接口的 `DEF_CFG_TPD` 方法, 以重置标定数据, 提高标定的准确性。

【Remarks: Before all calibrations (including blind element calibration, temperature measurement secondary calibration, pot lid calibration), please call the `DEF_CFG_TPD` method of the `restoreDefaultConfig` interface in the `IRCMD` class to reset the calibration data and improve the Calibration accuracy. 】

请注意: 调用该方法之后, 会重置所有的标定数据。【Note: After calling this method, all calibration data will be reset.】

即如果盲元标定结束之后, 在锅盖标定的时候, 调用了该方法, 则盲元标定的数据会丢失。

【That is, if this method is called when the lid is calibrated after the blind element calibration is completed, the data of the blind element calibration will be lost.】

4.1 准备工作【Preparation】

4.1.1 Step1

准备一台黑体，温度设置为T0。

【Prepare one black body, and set the temperature to T0.】

建议高低增益下分别设置应用场合的中值温度

【It is recommended to set the median temperature for the application scenarios under high and low gain respectively】

人体测温只需要在高增益模式下标定T0，建议设置为308K

【Creature Radiometry only needs to calibrate T0 in high-gain mode, and 308K is recommended】

4.1.2 Step2

将黑体放置于模组视场中心，黑体面距离模组镜头约0.25m。

【The black body can be placed in the center of the module field of view, and the black face is about 0.25m away from the module lens.】

4.1.3 Step3

等待黑体达到设定温度

【Wait for the blackbody to reach the set temperature】

实际黑体发射率不可能为1，需要修正。黑体面温度均匀稳定的区域至少需要占据画面中心9*9像素以上。

【The actual blackbody emissivity cannot be 100% and needs to be corrected. The area with uniform and stable temperature should occupy at least 9 x 9 pixels in the center of the image.】

4.2 模组温度稳定后，后进入二次标定单点标定流程

【After the module temperature is stabilized, the process is entered】：

4.2.1 Step1

将黑体置于画面中心处。

【Place the body in the center of the picture】

4.2.2 Step2

调用 `setPropAutoShutterParameter` 函数的 `SHUTTER_PROP_SWITCH` 为0，关闭自动快门。

【Call `setPropAutoShutterParameter` function with `SHUTTER_PROP_SWITCH` 0 to close the auto FFC.】

```
1  /**
2  * 设置自动快门参数【Set property page's auto shutter parameters】
```

```

3  * 需要 {@link #saveSpiConfig(int)} 的 {@link
CommonParams#SPI_MOD_CFG_PROPERTY_PAGE}才保存, 否则不保存
4  *
5  * @param shutter_param param during {@link CommonParams#SHUTTER_PROP_SWITCH
自动快门开关【Automatic shutter switch】},
6  *                                {@link CommonParams#SHUTTER_PROP_MIN_INTERVAL 最小快
门间隔【Minimum shutter interval】},
7  *                                {@link CommonParams#SHUTTER_PROP_MAX_INTERVAL 最大快
门间隔【Maximum shutter interval】}
8  *                                {@link
CommonParams#SHUTTER_PROP_TEMP_THRESHOLD_OOC},
9  *                                {@link CommonParams#SHUTTER_PROP_TEMP_THRESHOLD_B 自
动快门校正触发阈值【Automatic shutter correction trigger threshold】} ,
10 * @param value
11 * @return 0 is success
12 * @see #getPropAutoShutterParameter(int)
13 * @see #saveSpiConfig(int)
14 */
15 public int setPropAutoShutterParameter(int shutter_param, int value)

```

4.2.3 Step3

调用 `shutterUpdate` 函数, 进行快门校正

【Call `shutterUpdate` function for FFC】

```

1  /**
2  * 打快门【manual shutter】
3  * The firmware from 3.6 allow to shutter manually every 5 s
4  *
5  * @return 0 is success
6  */
7  public int shutterUpdate()

```

4.2.4 Step4

采集黑体的温度, 调用 `setTPDKtBtRecal1Point` 函数, 参数为黑体的实际开尔文温度 K(integer)。

【Collect blackbody data, call `setTPDKtBtRecal1Point` function interface, and set the parameter as the actual blackbody temperature Kelvin K(integer). The wait time for the USB command interface is about 11 seconds, and the wait time for the I2C command interface is about 20 seconds.】

```

1  /**
2  * 单点二次标定【Recalculate the kt_bt by 1 point's temperature.】
3  * need black body
4  * 标定时间约为5s, 可以保存【The calibration time is about 5s and can be
saved】
5  *
6  * @param temperature the target black body's temperature.
7  *                    units:K(integer)
8  * @return 0 is success
9  */
10 public int setTPDKtBtRecal1Point(int temperature)

```


4.2.5 Step5

调用 `setPropAutoShutterParameter` 函数的 `SHUTTER_PROP_SWITCH` 为1, 打开自动快门。

【Call `setPropAutoShutterParameter` function with `SHUTTER_PROP_SWITCH` as 1 to open the auto FFC.】

```
1  /**
2   * 设置自动快门参数【Set property page's auto shutter parameters】
3   * 需要 {@link #saveSpiConfig(int)} 的 {@link
CommonParams#SPI_MOD_CFG_PROPERTY_PAGE}才保存, 否则不保存
4   *
5   * @param shutter_param param during {@link CommonParams#SHUTTER_PROP_SWITCH
自动快门开关【Automatic shutter switch】},
6   *           {@link CommonParams#SHUTTER_PROP_MIN_INTERVAL 最小快
门间隔【Minimum shutter interval】},
7   *           {@link CommonParams#SHUTTER_PROP_MAX_INTERVAL 最大快
门间隔【Maximum shutter interval】}
8   *           {@link
CommonParams#SHUTTER_PROP_TEMP_THRESHOLD_OOC},
9   *           {@link CommonParams#SHUTTER_PROP_TEMP_THRESHOLD_B 自
动快门校正触发阈值【Automatic shutter correction trigger threshold】} ,
10  * @param value
11  * @return 0 is success
12  * @see #getPropAutoShutterParameter(int)
13  * @see #saveSpiConfig(int)
14  */
15 public int setPropAutoShutterParameter(int shutter_param, int value)
```

4.3 注意事项【Notes】

- 标定失败后, 需要调用 `restoreDefaultConfig` 接口的 `DEF_CFG_TPD` 重置温度计算参数, 重新进行二次标定

【If the calibration fails, call `DEF_CFG_TPD` of `restoreDefaultConfig` to restore the calibration data and perform secondary calibration again. 】

```
1  /**
2   * 恢复出厂设置,重置CFG参数【Restore factory settings, reset CFG
parameters】
3   * 立即生效, 可以保存【Effective immediately and can be saved】
4   *
5   * @param defaultCfgType {@link CommonParams#DEF_CFG_ALL 重置所有参数
【Reset all parameters】},
6   *           {@link CommonParams#DEF_CFG_TECLESS_K},
7   *           {@link CommonParams#DEF_CFG_TPD 重置测温参数
【Reset temperature measurement parameters】},
8   *           {@link CommonParams#DEF_CFG_PROP_PAGE},
9   *           {@link CommonParams#DEF_CFG_USER_CFG} ,
10  *           {@link CommonParams#DEF_CFG_DEAD_PIXEL 重置盲元
表【Reset the blind meta table】}
11  * @return 0 is success
12  */
13 public int restoreDefaultConfig(int defaultCfgType)
```

- 若模组为可切换高低增益版本，需要分别进行二次标定。

【If the module is a switchable high-low gain version, perform secondary calibration in two cases. 】

- 必须在开启测温功能的情况下采集黑体数据，这样才可以评估二次标定的效果。

【Data must be collected with temperature measurement enabled so that the effect can be evaluated.】